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RESEARCH ARTICLE

Revealing the Influence of Semantic Similarity on Survey Responses: A Synthetic Data Generation Approach

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This work involved human subjects or animals in its research. The authors confirm that all human/animal subject research procedures and protocols are exempt from review board approval.

ABSTRACT Questionnaires are essential for measuring self-reported attitudes, beliefs, and behaviour in many research fields. Semantic similarity of the questions is recognized as a source of covariance in the human data, implying that response patterns partly arise from the questionnaire itself. A practical method to assess the influence of semantic similarity could significantly facilitate the design of questionnaires and the interpretation of their results. The current study presents a novel method for estimating the influence of semantic similarity for questionnaires with Likert-scale responses. The method transforms Likert-scale responses into natural language sentences and applies the Sentence-BERT algorithm to compute a semantic similarity matrix. Synthetic response data are generated using the semantic similarity matrix and a noise parameter as input. Synthetic data can then be analysed using the same tools as human survey data, making the comparison straightforward. The method was tested with a questionnaire measuring the acceptance of automated driving. Synthetic data explained 40% of the observed correlations in the human response data. This means that semantic similarity substantially influenced responses. Using synthetic data, it was possible to identify the same factor structure as in the human data and to identify relationships between factors that might have been inflated by semantic similarity. This demonstrates that semantically generated synthetic data could help in designing multi-factor questionnaires and correctly interpreting the found relationships between factors.

INDEX TERMS Questionnaire, synthetic data, natural language processing, large language models, technology acceptance.

I. INTRODUCTION

Questionnaires can measure the self-reported attitudes, beliefs, and behaviour of human respondents. Responses are often given on a Likert scale, where respondents first read a statement (or prompt) and then indicate their agreement with the statement on a scale: Strongly disagree, Disagree, Neither agree nor disagree (or Neutral),

Agree, Strongly agree [1]. For the analysis, the responses are typically coded numerically (e.g. Strongly disagree = 1 at the lowest end, and Strongly agree = 5 at the highest end).

Numeric coding makes it possible to use various statistical tools to analyse the data. Confirmatory Factor Analysis (CFA) can be used for identifying latent factor structure, which combines multiple questions together, improving the validity and reliability of the measurement [2], [3]. Structural Equation Modelling (SEM) is typically applied to investigate

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the structural path relationships between the latent factors [3]. The results of these methods are influenced by the covariance in the response data.

Semantic similarity between questionnaire statements has been recognized as a significant source of covariance [4], [5]. Statements are semantically similar when they have similar meanings, even if they use different words or sentence structures. This implies that the factor structure and the relationships between factors might be partly determined by the design of the questions themselves. However, the questionnaire items measuring a latent construct may need to be semantically similar to some extent. Otherwise, they would not measure the same concept, and the factors would not achieve sufficient statistical validity and reliability. Problematic is, however, if the survey responses are determined mostly or solely by semantic similarity rather than by participants' attitudes, beliefs and behaviour. Currently, it is challenging to quantify the influence of semantic similarity on the survey responses e.g. on a Likert scale, as analysing semantic similarity requires advanced Natural Language Processing (NLP) tools.

The current study presents a novel method for estimating the influence of semantic similarity on survey responses for questionnaires with Likert-scale responses. The method estimates the semantic similarity of the Likert-scale responses and generates synthetic data reflecting it. Synthetic data can then be analysed using common analysis tools such as CFA and SEM, and the results compared to those obtained with human data.

The method posits that the survey responses to Likert-scale items could be represented as sentences of natural language instead of numerically coding them. When the responses are natural language, it is possible to apply computational algorithms to compare the semantic similarity of the responses, that is, how close the responses are in meaning.

The responses are reconstructed from the questionnaire statements and their response options by producing sentences with a template "I <response> that <statement>". For example, "I agree that I would use an automated car." or "I strongly disagree that I would trust automated cars." After constructing the responses, they are transformed into sentence embeddings using a large language model (LLM) based Sentence-BERT (SBERT) algorithm [6], which is a modification of the BERT (Bidirectional Encoder Representations from Transformers). Sentence embeddings are numeric vectors representing the content of sentences. As numeric vectors, their similarity can be compared using e.g. cosine similarity. The result of the process is a semantic similarity matrix¹ between all the possible responses within a questionnaire. Then the semantic similarity matrix is used to generate

synthetic response data, where semantically similar responses are preferred over semantically dissimilar.

We tested the method in a case study with a questionnaire focusing on the acceptance of conditionally automated cars [7]. We generated synthetic response data based on the questionnaire and compared it to the human data in terms of covariances (between variable correlations), measurement model (factor structure) and structural model (relationships between factors).

The aim of the present study was to demonstrate how semantically generated synthetic data could be used for designing surveys which have the intended measurement model, and where the relationships between the factors are not inflated by the semantic similarity of the response options. Being able to test the questionnaires before collecting any data would improve the quality and effectiveness of the research. Synthetic data could also help in interpreting the structural relationships in the data already collected. Because the synthetic data can be analysed using the same tools as the human response data, its use is straightforward.

To the best of our knowledge, the proposed method of representing the Likert-scale responses as natural language sentences, comparing them for their semantic similarity, and using the resulting similarity matrix to generate synthetic data is novel and has not been presented in prior research.

II. BACKGROUND

A. SEMANTIC SIMILARITY AND SURVEY RESPONDING

The questionnaire used in the case study was based on an extended version of the Unified Theory of Technology Acceptance (UTAUT), namely UTAUT2 [8]. UTAUT2 suggests that the actual use of technology is predicted by behavioural intention to use it, which itself is influenced by factors such as performance expectancy, facilitating conditions, social influence, and hedonic motivation. UTAUT and UTAUT2 have been widely applied, also in the context of automated driving [9], [10], [11], [12].

UTAUT factors are often found to be strongly positively correlated [13]. This has raised the question whether the factors are independent or just reflecting a single general acceptance factor [14]. On the other hand, de Winder and Nordhoff [15] suggested that the individuals' response styles might be the underlying reason for the high positive correlations.

Arnulf et al. [4] proposed the semantic theory of survey responding (STSR) which essentially states that a substantial part of the covariance between the questionnaire items can be explained by their semantic similarity. This means that the factor structure and the relationships between the factors found in human response data may not only reflect the implied constructs (e.g., attitudes, beliefs or behaviours), but they partly rise from the way how humans use language to represent concepts [16]. Semantic similarities might be another source of strong positive correlations between factors in UTAUT and UTAUT2-based questionnaires.

¹The similarity matrix can be also calculated using the numeric values, too, but the advantage of the semantic similarity matrix is that it can represent similarity in the meaning of the whole response (statement and response option), while the one based on numerically coded values could only represent similarity of the numerically coded response options.

Previous studies on semantic similarity and survey responding have focused on the semantic similarity of the questionnaire statements. Arnulf et al. [4] showed that in the Multifactor Leadership Questionnaire the semantic similarity of the questionnaire statements replicated the intended factor structure and explained up to 86% of the covariance between the factors in the human data. Gefen et al. [17] showed that lexical closeness could replicate the measurement model of the Technology Acceptance Model (TAM) [18] using two independent semantic spaces derived from business journal articles and news articles. They also showed that lexical closeness could partly replicate the structural model of TAM applied in another study. Semantic similarity of the statements has been also applied for predicting responses to questionnaires based on the semantic similarity of the statements using general linear models [5]. Arnulf et al. [16] provide a comprehensive review of the studies.

Analysis of semantic similarity may help in designing surveys. Semantically similar statements are more likely to load on the same factor [17]. Semantic similarity of questionnaire statements can be even viewed as necessary for arguing that the statements would measure the same constructs. On the other hand, if the items intended to measure different constructs are semantically too similar, the resulting correlations may create difficulties in separating the constructs, decreasing the discriminant validity. The inflated relationships may also suggest connections which are purely semantic. In the extreme case, the design of the questionnaire itself could determine the results of the study. Therefore, the survey researchers need practical tools which make the potential influence of semantic similarity visible [4], [16].

B. SBERT AND SEMANTIC SIMILARITY

Previous studies on the influence of semantic similarity on survey responses have used non-LLM-based methods, such as Latent Semantic Analysis (LSA) [19]. LSA uses singular value decomposition to represent text in a high-dimensional space based on word co-occurrences, measuring semantic similarity in terms of lexical proximity while disregarding word order, syntax, and morphology.

In the current study, we use SBERT [6]. SBERT is a computationally efficient modification of the BERT architecture. Variants of BERT have attained state-of-the-art performance across numerous NLP tasks [20]. SBERT offers several advantages over LSA. SBERT utilizes LLMs to capture the meaning of whole sentences, providing a more nuanced understanding of the text.

Contemporary LLMs, including BERT and SBERT, are Transformer models pre-trained on generic linguistic tasks. Transformers are deep neural networks that use attention mechanisms to process sequential data, such as sentences, without recurrent connections [21]. Transformers produce a separate contextual encoding for each token, such as words, in parallel. Because of this, further means are needed for representing full sentences. BERT represents full sentences

by using a separate <CLS> token in front of every input. The contextual embedding of this token represents the entire sentence. SBERT instead takes all the contextual encodings and combines them via e.g. averaging or max pooling.

Both BERT and SBERT can represent the semantic content of two sentences as vectors, which are called sentence embeddings. Semantic similarity of two sentences can then be compared in terms of an objective function such as cosine similarity. Cosine similarity is the cosine of the angle between two vectors in the vector space. It is calculated as a dot product of the vectors $\mathbf{A}$ and $\mathbf{B}$ divided by the product of their length (Eq. 1). Two vectors with the same direction have a cosine similarity of 1, two orthogonal vectors have a similarity of 0, and two opposite vectors have a similarity of -1 . Cosine similarity does not depend on the magnitude of the vectors. Cosine similarity becomes equivalent to Pearson correlation if the vectors are mean-centred.

$$\cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \cdot \|\mathbf{B}\|} \quad (1)$$

A major point of contention is how the sentence embeddings should be interpreted semantically [22], [23]. One prevalent hypothesis is that Transformers capture distributional semantics, where the meaning of a word is determined by its relations to other words in the language [24], [25], [26]. In this interpretation, Transformers would expand on prior distributional representation schemes like LSA, by taking into account word order, inflection, and other contextual information beyond mere lexical co-occurrence patterns. This interpretation is especially salient for BERT's standard pre-training task of masked token prediction, where the identity of a token is predicted based on its surrounding context [27].

The training data of SBERT is drawn from corpora labelled by human annotators representing semantic links between sentences, such as whether they stand in an entailment or paraphrase relation. This makes its interpretation more straightforward in terms of semantics. SBERT's embedding representations track semantic similarity as characterized in its labelled training data (such as paraphrase relations), because this is what SBERT is specifically trained to do.

LLMs often fail at recognizing negation. Models have shown a tendency to ignore negation in masked token prediction [28], generate incorrect answers to negated prompts [29], and struggle to recognize entailment or contradiction relations involving negated sentences [30], [31]. Surprisingly, increasing model size does not improve performance with negated data; it can even deteriorate this [29]. It thus seems that understanding negation does not automatically emerge from the pre-training of LLMs, in contrast to many other linguistic capacities [32]. Surveys often include also negated statements, because they can be used in addressing the effects of response styles, such as the tendency to select one end of the scale [2]. Therefore, negated responses need special attention with SBERT.

The original Transformer is an encoder-decoder network, but BERT and its derivatives, including SBERT, are only encoder models, which means that they are built for analyzing their input rather than generating text. In recent years, Generative Pretrained Transformers (GPT), which are encoder-decoder models, have demonstrated remarkable capabilities in generating coherent textual responses based on the given prompt. They can be even prompted to respond to a questionnaire in a way that the response patterns show similarities to human respondents [39], [40]. GPT models are under intensive development, and they are also progressing in understanding the negation [33]. Recent work has presented SGPT (GPT Sentence Embeddings for Semantic Search) [34], where the capabilities of very large GPT models could be used in a similar way to SBERT. While decoder LLMs tend to dominate contemporary NLP applications, SBERT remains a top-performing sentence embedding technique [35]. It can be regarded as the most dominant contemporary way of estimating the semantic similarity of two sentences. Trained and tested SBERT models are also freely available via the huggingface.com website. For the above reasons, the current study leveraged SBERT to analyse semantic similarities in a UTAUT2 questionnaire to evaluate the influence of semantic similarities on the survey responses.

III. METHODS

We used a subset of a questionnaire measuring acceptance of automated driving for which also empirical survey data was available. The semantic similarity of the responses was analysed and synthetic response data were generated. Human and synthetic data were compared in terms of their covariances, and measurement and structural models. The workflow is summarized in Fig. 1.

A. QUESTIONNAIRE AND HUMAN DATA

The questions analysed in the current study are listed in Table 1. The responses were given on a scale: Strongly disagree, Disagree, Neutral, Agree, Strongly agree. The items were intended to measure the five of the UTAUT2 factors, namely behavioural intention (BI), performance expectancy (PE), social influence (SI), hedonic motivation (HM), and facilitating conditions (FC), and trust as an additional factor (TRU).

The empirical survey data with the questionnaire were collected as a part of the L3Pilot project (<https://l3pilot.eu/>). The data collection was performed by a market research company via their online panels in nine countries (France, Hungary, Germany, China, Brazil, Japan, Russia, USA, UK). The questionnaire was translated into the official/majority language of the country. In total, 9,339 responses were collected (> 1,000 per country). The participants gave an informed consent to participate. The data collection is described in Nordhoff et al. [7]. The data is openly available in Zenodo [36].

B. SEMANTIC SIMILARITIES FOR LIKERT-SCALE RESPONSES

We used a template “I <response> that <statement>” to generate responses from the questionnaire statements and their response options. For example, with response option “Agree” to the statement “I plan to use a conditionally automated car once it becomes available” would become “I agree that I plan to use a conditionally automated car once it becomes available.” After the conversion, the semantic similarities between all the responses were calculated using SBERT [6].

The Python module SentenceTransformer (v. 2.2.2) was used. It provides open access to thousands of SBERT variants that differ in detail concerning training data, model

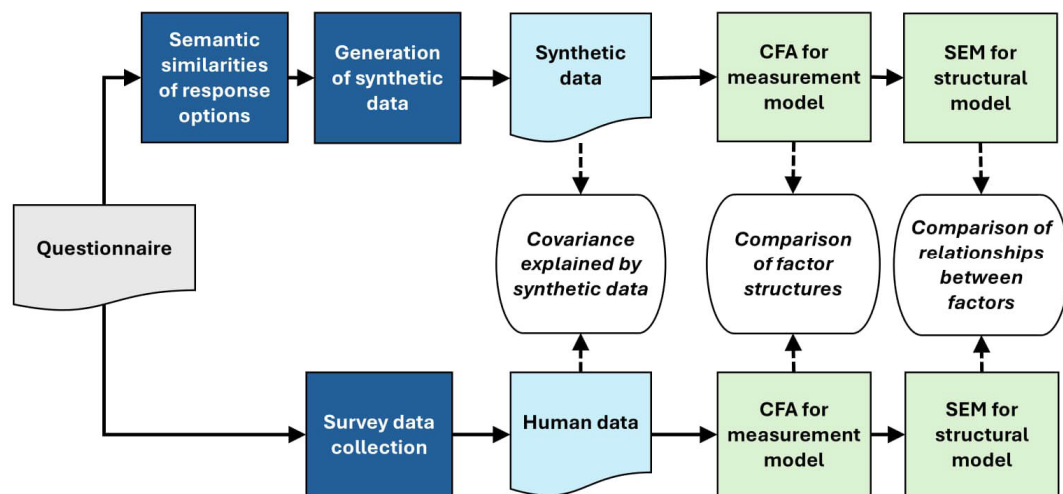


FIGURE 1. Workflow of the study. Synthetic data were generated based on the semantic similarities and analysed (upper path). The same analyses were done for previously collected survey data (lower path). The results from the two datasets were compared (middle).

TABLE 1. Questionnaire items analysed with their IDs, intended constructs, and questionnaire statements. The IDs correspond the variable names in the openly available data [36].

ID	Latent construct	Statement
q12.1	TRU	I would be suspicious of conditionally automated cars.
q12.2	TRU	I would trust conditionally automated cars.
q12.3	TRU	I would engage in other tasks while the conditionally automated car is in control.
q12.4	TRU	I would feel hesitant about using a conditionally automated car.
q12.5	BI	I plan to use a conditionally automated car once it becomes available.
q15.1	BI	I intend to use a conditionally automated car in the future.
q15.2	BI	The next car I buy will be a conditionally automated car, if it is available.
q14r.1	PE	Using a conditionally automated car would help me reach my destination more safely.
q14r.2	PE	Using a conditionally automated car would help me reach my destination more comfortably.
q14r.3	PE	The cost of the conditionally automated car would be the most important thing I would consider before purchasing one.
q14r.4	PE	The benefits of using a conditionally automated car would be the most important thing I would consider before purchasing one.
q14r.5	PE	Using a conditionally automated car would help me to reach my destination faster.
q14r.11	SI	I assume that people whose opinions I value would prefer that I use a conditionally automated car.
q14r.12	SI	Using a conditionally automated car would give me status and prestige among people important to me.
q14r.13	SI	It would make me proud to own a conditionally automated car.
q14r.14	HM	Using a conditionally automated car would be enjoyable.
q14r.15	HM	I find it important that the conditionally automated car has a sleek and cool design.
q14r.16	HM	The brand of the conditionally automated car would be the most important thing I would consider before purchasing one.
q14r.17	FC	I could acquire the necessary knowledge to use a conditionally automated car.
q14r.18	FC	I would be able to get help from my friends and/or family when I have difficulties using a conditionally automated car.
q14r.19	FC	I would not be able to get help from car dealers when I have difficulties using a conditionally automated car.
q14r.20	FC	I would be able to read the driver manual when I have difficulties using a conditionally automated car.

size, pooling functions, and other hyperparameters. Five models were tested (Appendix A). The final model was selected based on its ability to produce synthetic data where the between-variable correlations matched the observed between-variable correlations in the human data. all-mpnet-base-v2 model gave the best match and it was used.

SBERT produced sentence embeddings, representing the semantic content of the response. Semantic similarity was calculated as the cosine similarity of the sentence embeddings [6]. Semantic similarity calculations were run using Python version 3.11. The resulting semantic similarity matrix of the responses was then used to generate synthetic data.

An example of the cosine similarities between responses to two different statements is given in Table 2. The semantic similarities for the Likert-scale responses demonstrated two desired features: 1) Responses to two related statements had a higher semantic similarity compared to responses to an unrelated statement. 2) The closer the

two responses were on the scale, the higher their semantic similarity. This suggests that the semantic similarities between the responses may reflect the correlations in human responses.

Two statements (q12.1, q12.4, q14r.19) were stated negatively and the answers to them were intended to be reverse coded. Despite SBERT considering the whole sentence, it did not perform well in recognizing the negation in the sentences. SBERT gave the semantic similarity estimates as if the sentences would not have been negated, as exemplified in Table 3. This was unsurprising, because LLMs are known not to perform well with negated sentences [28], [29], [30], [31].

We also calculated the semantic similarities for the statements only, that is, without the response option. An example of the semantic similarities between questionnaire statements is shown in Table 4. Semantic similarities between statements intended for the same construct were higher than statements intended for different constructs.

TABLE 2. Example of cosine similarities of textually coded responses for two questions (q15.1 and 14r.1).

q14r.1: Using a conditionally automated car would help me reach my destination more safely.		Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
q15.1: I intend to use a conditionally automated car in the future.	Strongly disagree	0.87	0.83	0.78	0.72	0.72
	Disagree	0.85	0.86	0.80	0.76	0.75
	Neither agree nor disagree	0.77	0.78	0.88	0.73	0.74
	Agree	0.65	0.68	0.69	0.87	0.88
	Strongly agree	0.64	0.65	0.68	0.85	0.88

TABLE 3. Example of cosine similarities of textually coded responses for two questions, one with an affirmative (q15.1) and another with a negated (q12.4) statement.

q12.4: I would feel hesitant about using a conditionally automated car.		Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
q15.1: I intend to use a conditionally automated car in the future.	Strongly disagree	0.87	0.83	0.80	0.78	0.81
	Disagree	0.85	0.84	0.80	0.77	0.79
	Neither agree nor disagree	0.79	0.77	0.87	0.72	0.76
	Agree	0.72	0.74	0.75	0.82	0.84
	Strongly agree	0.70	0.70	0.74	0.79	0.83

TABLE 4. Example of semantic similarities between statements intended for different latent constructs.

Question ID and intended construct	q12.5 (BI)	q15.1 (BI)	q14r.1 (PE)	q14r.2 (PE)	q14r.15 (HM)
q12.5 (BI)	1.00	0.96	0.81	0.83	0.74
q15.1 (BI)	0.96	1.00	0.82	0.84	0.77
q14r.1 (PE)	0.81	0.82	1.00	0.98	0.69
q14r.2 (PE)	0.83	0.84	0.98	1.00	0.70
q14r.15 (HM)	0.74	0.77	0.69	0.70	1.00

C. GENERATION OF SYNTHETIC RESPONSE DATA

We created a semantic response agent to generate synthetic data. The response agent was programmed in the R language. The code is published in Zenodo [37]. The agent had only two parameters: the semantic similarity matrix of the questionnaire responses and the amount of noise added to its responses.

The agent aims to be semantically coherent in its responses in a greedy way, by choosing the response which is closest in meaning to any of its previous responses. In other words, it takes all the five possible response sentences to the current

question and picks up the one which has the highest semantic similarity to any of the previous response sentences. The first question is answered by random because there are no previous responses to compare with.

The algorithm had the following steps:

1. The order of the questions to be answered is first randomized.
2. The agent answers the first question by randomly picking a response sentence for the question.
3. The randomly picked response sentence is stored in the agent's memory.

4. The agent moves to the next question. The possible response sentences to the question are compared to the previous response sentences stored in its memory. The response sentence which has the highest semantic similarity to any of the memorized response sentences is chosen as the answer.
5. The chosen response sentence is added to the agent's memory.
6. The agent moves back to step 4 until all the questions are answered.

The possibility to add noise to the generated responses was also implemented. Noise values were drawn from Gaussian distribution with $\mu = 0$ and varying σ . The noise was discretized by rounding to the nearest integer. Noise was then added to the numerically coded response value. Because the resulting value could be outside the numerical scale of the responses, the final response value was floored to be at least one or ceiled to be at most five. In other words, the noise could shift the response either toward stronger agreement or stronger disagreement on the Likert scale. Then the response sentence corresponding to the final numerical value was retrieved, used as an answer, and stored in the agent's memory (see Appendix B for an example).

The semantic similarity matrix of the response sentences was used to generate 1,000 responses with different magnitudes of noise ($\sigma = 0, 0.5, 1.0, 2.0, 3.0$). 1,000 responses were deemed enough for investigating the measurement and structural models in the data.

The noise parameter affected the within-agent standard deviation. The within-agent standard deviation in the generated data were 0.65 with $\sigma = 0$, 0.85 with $\sigma = 0.5$, 1.09 with $\sigma = 1.0$, 1.44 with $\sigma = 2.0$, and 1.61 with $\sigma = 3.0$. The corresponding within-respondent standard deviation in the human data was 0.89, suggesting the synthetic data with some amount of noise resembles more human data in this respect.

D. DATA ANALYSIS

Human and synthetic data were compared using the commonly used methods, which rely on numeric coding of the responses. The following numeric coding was used: Strongly disagree = 1, Disagree = 2, Neutral = 3, Agree = 4, Strongly agree = 5.

Pearson correlations between all the numerically coded responses were calculated using the numerically coded data. 22 questions generated a correlation matrix of 484 items, of which 22 were self-correlations. This resulted in six correlation matrices, one for the observed human data, and one for each synthetic dataset with different amounts of noise.

We then used linear regression models to examine how well the correlations in the synthetic data explained the correlations in the human data. After excluding the self-correlations, there were 462 pairs of correlations for each of the regression models.

Because previous research has focused on the statements and their semantic similarity, we also ran a linear regression

model where the observed correlations were regressed to the semantic similarities between statements only.

The human and synthetic data were then further compared. We performed CFA with both datasets to identify the measurement model. We also ran SEM to investigate the structural model between the factors and to compare the results between the human and synthetic data.

CFA was performed using package lavaan (v. 0.6-17), which gives the widely used model fit statistics. Cronbach alphas were calculated using the package psych (v. 2.4.1) and average variance extracted (AVE) with the package semTools (v. 0.5-6).

For SEM, a Bayesian version of lavaan package, blavaan (v. 0.5-5) was used. blavaan gives the posterior distributions for the estimated path coefficients, which makes it straightforward to compare coefficients between human and synthetic datasets.

Data analyses and generation of synthetic response data were performed in R version 4.3.3.

IV. RESULTS

A. SYNTHETIC AND HUMAN DATA

Pearson correlations between the questionnaire items were calculated within the human and synthetic datasets. The synthetic dataset with a moderate amount of noise ($\sigma = 1.0$) explained the highest proportion of variance in the human data (40%) (Table 5). The relationship between the synthetic data with the highest explained variance and the human data is visualised in Fig. 2.

Apparently, some noise in the responses generates a more varied pool of memorized response sentences, providing better matches to the following responses. With too much noise, the influence of semantic similarity started to decrease.

The correlations for the synthetically generated data were calculated without first reverse coding the responses given by the agent. SBERT was not able to differentiate between positively and negatively phrased response options but gave similarities as if the sentence was not negated (for an example, see Table 3). Consequently, the responses given by the semantic response agent showed the pattern as if the response data would have been already reverse coded.

For comparison, we also calculated how well the semantic similarities of the questionnaire statements alone could explain the observed correlations (Fig. 3). The linear regression model showed that 40% of the variance was explained (Table 6).

B. MEASUREMENT MODEL

In the following, we will demonstrate how the semantically generated synthetic data can be used to identify potential problems in the measurement model (i.e. factor structure).

Factors are typically chosen according to the used theoretical framework and the corresponding indicators based on the literature or created from scratch. Then the factors are tested

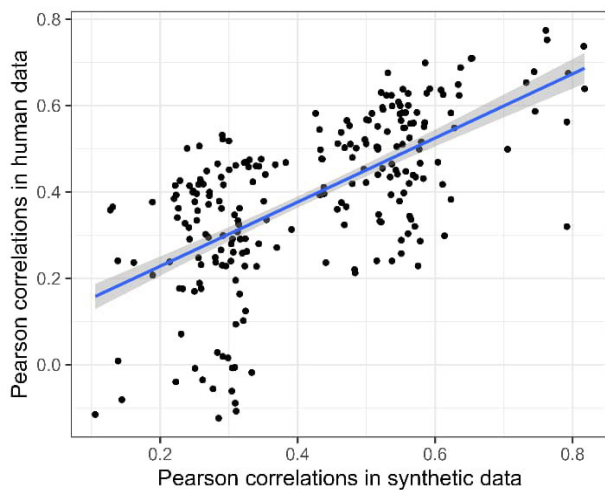


FIGURE 2. Between-item correlations in synthetic response data (x-axis) and human response data (y-axis). Linear regression line with 95% confidence intervals added.

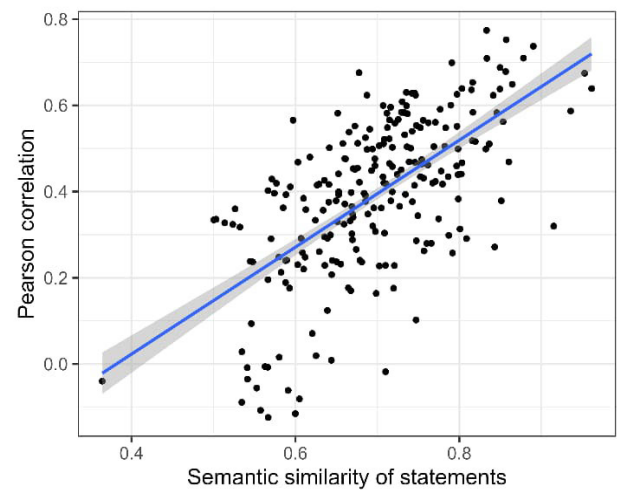


FIGURE 3. Relationship between the semantic similarity of the questionnaire statements (x-axis) and between-item correlations in the human response data. Linear regression line with 95% confidence intervals added.

for their validity and reliability. The fit of the measurement model is confirmed using CFA.

The current questionnaire was designed based on the UTAUT2 model. We set out to identify UTAUT2 factors Behavioural Intention (BI), Performance Expectation (PE), Facilitating Conditions (FC), and Social Influence (SI). In addition, we included a factor for Trust (TRU).

Analysis of the data revealed that all the indicators did not load perfectly to their designated factors. For demonstration purposes, we selected the indicator variables for each factor based on an exploratory network analysis of the

indicator variables in the human data. Based on the network analysis, it is easy to recognize which variables form correlated clusters of variables. It can also show if two indicator variables are so highly correlated that they might create a problem.

Six factors were formed and tested for their reliability with Cronbach alphas (Table 7). In the synthetic data, four out of six factors had sufficient reliability and validity. Cronbach alphas exceeded the commonly used threshold of 0.70 for BI, PE, SI and TRU, but not for FC and HM. In the human data,

TABLE 5. Linear regression model predicting the correlations in the human data by the correlations in the synthetic datasets with different amount of noise. R^2 represents the proportion of the variance in the dependent variable which can be explained by the independent variables of the model.

	Dependent variable: Correlation in human data				
	Syn. data $\sigma = 0$	Syn. data $\sigma = 0.5$	Syn. data $\sigma = 1.0$	Syn. data $\sigma = 2.0$	Syn. data $\sigma = 3.0$
Correlation in synthetic data	0.400*** (0.040)	0.575*** (0.041)	0.742*** (0.042)	1.023*** (0.064)	1.192*** (0.090)
Constant	0.091** (0.031)	0.032 (0.027)	0.080*** (0.019)	0.214*** (0.013)	0.283*** (0.011)
Observations	462	462	462	462	462
R^2	0.182	0.297	0.400	0.360	0.275
Adjusted R^2	0.180	0.295	0.398	0.359	0.273
Residual Std. Error (df = 460)	0.168	0.155	0.144	0.148	0.158
F Statistic (df = 1; 460)	102.394***	194.145***	306.341***	259.091***	174.289***

Note:

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

TABLE 6. Linear regression model predicting the correlations in the human data by the semantic similarities of the questionnaire statements only. R^2 represents the proportion of the variance in the dependent variable which can be explained by the independent variables of the model.

	Dependent variable: Correlation in human data:
Semantic similarity of statements	1.241 ^{***} (0.071)
Constant	-0.473 ^{***} (0.050)
Observations	462
R^2	0.399
Adjusted R^2	0.397
Residual Std. Error (df = 460)	0.144
F Statistic (df = 1; 460)	304.817 ^{***}

Note:

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Cronbach alphas for BI, PE, SI and TRU similarly exceeded the threshold.

The model with four factors (BI, PE, SI and TRU) was entered into CFA to evaluate the fit. Commonly used thresholds for acceptable model fit are $CFI \geq 0.95$, $TLI \geq 0.95$, $RMSEA \leq 0.50$, $SRMR \leq 0.80$ [3], [38].

With the synthetic data, X^2 test indicated that the fit was not perfect, but that was expected considering the relatively large sample size ($X^2(38) = 182,711$, $p < 0.001$). Other fit indices showed acceptable fit ($CFI = 0.983$, $TLI = 0.976$, $RMSEA = 0.062$, 90% CI [0.53, 0.71], $SMRM = 0.030$).

CFA was run with the same measurement model using the human data. Again, X^2 test showed that the model did not fit perfectly ($X^2(38) = 1902.133$, $p < 0.001$). The other fit indices showed acceptable fit ($CFI = 0.973$, $TLI = 0.961$, $SRMR = 0.025$) or close to acceptable fit ($RMSEA = 0.072$, 90% CI [0.70, 0.65]).

The average variance extracted (AVE) was above the recommended threshold of 0.50 for all the factors in both models (Table 8). Absolute residual correlations were less than 0.09 for all indicators in both models.

In summary, the same measurement model could be identified with synthetic and human data. This demonstrates that synthetic data can be a useful tool for testing the measurement model even before collecting data.

C. STRUCTURAL MODEL

Synthetic data can be used to analyse the influence of semantic similarity in structural models (i.e. relationships between factors). For the sake of example, we tested a model investigating the role of TRU as a mediator of PE and SI on BI. Fig. 4 shows the standardized path estimates between the factors based on synthetic data and human data, and their difference.

The structural model based on human data had relatively strong paths (> 0.40) from SI to BI, from PE to TRU, and from TRU to BI. PE and SI had a strong covariance (0.90). The direct effect of PE on BI was weak (0.085), but the mediated effect via TRU was stronger (0.26) (Table 9). The mediated effect of SI on BI via TRU was weak (0.074).

The interpretation of the structural model would be that positive performance expectations (PE) increase the intention to use automated cars (BI) by increasing the trust (TRU) in them. In addition, support or pressure from other people to use AVs (SI) also increases the intention to use automated cars (BI).

The same model was fitted with the synthetic data. If we interpret the path coefficients in the synthetic data model to reflect the influence of semantic similarity and assume that this is still fully present in the empirical data, it would be possible to estimate the residual effect in the empirical response data by subtracting the synthetic path coefficients from the empirical ones. It could be argued that the residual represents the influence of the semantic similarity of the responses.

The structural models were Bayesian SEMs. This means that it is possible to subtract the samples from the posterior distribution of the synthetic data model from the samples of the human data model to come up with the posterior distribution of their difference. Table 9 shows the mean and 95% credible intervals for the parameters in both models and their differences.

The paths from PE to BI and SI to TRU become negative after removing the semantic similarity effect. The effect of SI on BI vanishes to zero (zero was within the 95% credible interval). The paths from PE to TRU and from TRU to BI become weaker than in the human data. Covariance of PE and SI becomes weak.

TABLE 7. Tested factors and their Cronbach alphas. Values between zero and one. A higher Cronbach's alpha indicates greater internal consistency reliability, meaning the indicators measure the same latent construct.

Latent variable	Indicator variable	Synthetic data	Human data
		Cronbach alpha	Cronbach alpha
BI	q12.5	0.91	0.90
	q15.1		
	q15.2		
PE	q14r.1	0.92	0.84
	q14r.2		
	q14r.5		
FC	q14r.17	0.53	0.63
	q14r.20		
	q14r.11		
SI	q14r.12	0.81	0.82
	q14r.13		
	q14r.15		
HM	q14r.16	0.36	0.59
	q12.1		
TRU	q12.2	0.78	0.73

TABLE 8. Factorial structure identified with CFA in the synthetic and human data. Standardised loadings for each indicator. Loadings are given between minus one and one. Loadings closer to one mean a stronger positive relationship between an indicator and a factor. Average variance extracted (AVE) measures what proportion of variance in the factors the indicator variables explain (convergent validity), AVE values are between zero and one, where one is better.

Latent variable	Indicator variable	Synthetic data			Human data		
		Std. est.	SE	AVE	Std. est.	SE	AVE
BI	q12.5	0.901	0.008	0.781	0.855	0.003	0.755
	q15.1	0.898	0.008		0.881	0.003	
	q15.2	0.854	0.010		0.870	0.003	
PE	q14r.1	0.914	0.008	0.789	0.834	0.004	0.635
	q14r.2	0.864	0.010		0.792	0.005	
	q14r.5	0.884	0.009		0.767	0.005	
SI	q14r.11	0.667	0.020	0.612	0.739	0.005	0.610
	q14r.12	0.838	0.013		0.747	0.005	
	q14r.13	0.832	0.013		0.847	0.004	
TRU	q12.1	0.901	0.012	0.793	0.648	0.007	0.589
	q12.2	0.879	0.012		0.868	0.006	

This suggests that the effect of SI on BI in the human data can be due to the semantic similarity of the responses. In addition, the semantic similarities may have made PE and SI strongly covary. Semantic similarities may have also made the paths from PE to TRU and from TRU to BI stronger. If some other, less semantically similar indicator variables had been selected, the effect might not have been found in the human data. The mediation of PE on BI via TRU was not largely affected after removing the semantic similarity effect.

V. DISCUSSION

The current study presented a novel method for estimating the influence of semantic similarity for questionnaires with Likert-scale responses. The responses were first constructed from the statements and their response options (e.g. "I agree that I would use an automated car."). Semantic similarity was then estimated using an LLM-based SBERT algorithm [6]. The resulting semantic similarity matrix was used to generate synthetic response data. The method was applied

to a questionnaire measuring the acceptance of automated driving.

The synthetic response data showed a substantial correspondence to the observed human data. Synthetic data explained 40% of the covariance in the human data. We demonstrated that a measurement model (i.e. factor structure) fit for the human data could be identified with the synthetic data.

We also demonstrated the use of synthetic data for analysing the structural model using SEM. The synthetic and human data showed substantial differences in their structural models. This means that the results of the structural equation modelling could not be explained only by the semantic similarity of the response options in the studied questionnaire. Still, large path coefficients in the structural model based on the synthetic data suggest that some of the relationships in the human data might be due to the semantic similarity of the response options. In particular, the current research suggests that semantic similarities might contribute to the high correlations between factors in UTAUT-based questionnaires [13], [15].

Semantic similarity of the statements only could similarly predict 40% of the variance in the current questionnaire. This is in line with the previous studies which have focused on the semantic similarity of the statements and their influence on the response patterns [4], [5], [16], [17].

Recently generative LLMs have shown remarkable success in generating responses to surveys [39], [40]. The current results suggest that a part of their success might be due to LLMs' ability to capture the semantic similarities of the questionnaires themselves.

A. IMPLICATIONS FOR SURVEY RESEARCH

The current results show that semantic similarities of the questionnaire items may influence response patterns. The influence of semantic similarity means that survey responses are not pure reflections of respondents' attitudes, beliefs or behaviours. In addition, they reflect the relationships between concepts as they are used in human language, what Arnulf et al. [16] call a semantic grid. The semantic grid represents the mutual meaning between concepts, which may or may not reflect the true state of the world. On the one hand, the semantic grid makes it possible to do survey research with latent factors. Without the semantic grid, it would be challenging to come up with statements that are related to each other in a meaningful way.

The proposed method could be applied for three purposes in survey research: First, the analysis of semantic similarities can be a potent tool for identifying statements that group into latent constructs [16]. In practice, generating synthetic response data from the similarity matrix of responses could be used to check if a desired measurement model can be identified with the questionnaire even before collecting any

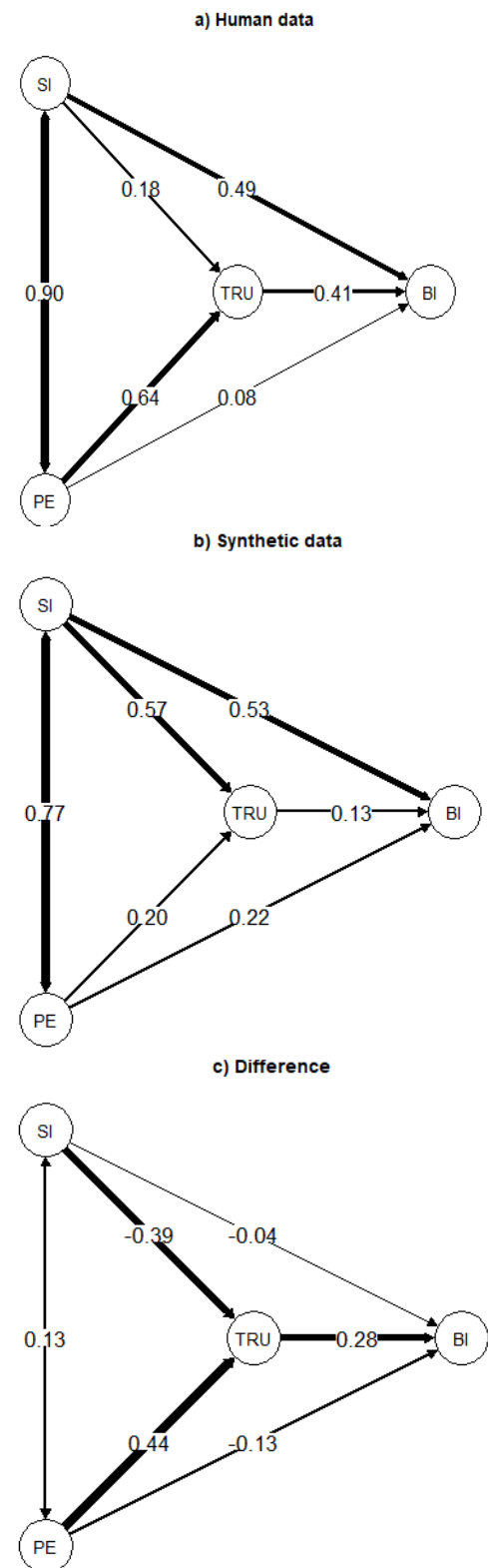


FIGURE 4. SEM based on a) human data, b) synthetic data and c) their difference.

data. This could help to ensure the internal consistency of the factors.

TABLE 9. Standardized estimates with human response data, synthetic data, and their difference. Means and 95% credible intervals (CI) for the posterior probability distributions are given. CIs are the highest-density intervals. Estimates are between minus one (perfect negative relationship) via zero (no relationship) to one (perfect positive correspondence).

Parameter	Human data		Synthetic data		Difference	
	Mean	95% CI	Mean	95% CI	Mean	95% CI
PE $\rightarrow$ TRU	0.64	[0.57, 0.70]	0.21	[0.10, 0.31]	0.44	[0.32, 0.57]
SI $\rightarrow$ TRU	0.18	[0.11, 0.24]	0.57	[0.47, 0.67]	-0.39	[-0.52, -0.28]
PE $\rightarrow$ BI	0.085	[0.028, 0.14]	0.22	[0.13, 0.30]	-0.13	[-0.23, -0.03]
SI $\rightarrow$ BI	0.49	[0.45, 0.54]	0.53	[0.43, 0.64]	-0.039	[-0.15, 0.074]
TRU $\rightarrow$ BI	0.41	[0.38, 0.44]	0.13	[0.055, 0.21]	0.28	[0.20, 0.36]
Covariance of PE and SI	0.90	[0.89, 0.91]	0.77	[0.73, 0.80]	0.13	[0.094, 0.17]
Mediation effect PE $\rightarrow$ TRU $\rightarrow$ BI	0.26	[0.23, 0.30]	0.026	[0.0057, 0.05]	0.24	[0.19, 0.28]
Mediation effect SI $\rightarrow$ TRU $\rightarrow$ BI	0.074	[0.047, 0.10]	0.076	[0.032, 0.10]	-0.0023	[-0.054, 0.052]

Second, testing the hypothesized structural model with the synthetic data could help in identifying relationships which might become inflated due to semantic similarity. Avoiding strong semantic similarities between items intended to measure two different constructs increases the chances that the found relationship is true. This would ensure the discriminant validity of the measurement.

Third, comparing the structural models based on the human data and the semantically generated synthetic data could be a way to identify in what degree the found relationships are semantical and in what degree due to the respondent's attitudes, beliefs or behaviours.

All in all, semantically generated synthetic data could help in designing surveys which measure what they are intended to measure and make researchers more aware of the potential impact of semantic similarity on response patterns. In the present study the method was applied to Likert-scale responses, but in principle, it could also be applied to other response formats if the statements and response options can be represented as natural language sentences.

B. LIMITATIONS

The current study has several limitations. First, the study did not test the generation of synthetic data with multiple questionnaires. Therefore, it is not possible to generalize to other questionnaires that 40% of the covariance could be explained by the semantic similarities.

Second, the study used a template to convert the responses into sentences. The template used may influence the estimated semantic similarity of the responses and thus influence the results.

Third, when estimating the semantic similarities between the responses, it was found that SBERT did not recognize negations in the questionnaire but treated negated sentences

as affirmed. The behaviour was in line with prior findings [28], [29], [30], [31], [32]. To compensate for this, we did not reverse code the items with negated statements in the synthetic data before analyses. While this fix worked with the current questionnaire, its applicability with other questionnaires should be carefully considered. For future work, this illustrates a need to focus on enhancing model performance specifically with negated sentences. Possible methods include synthetic data augmentation [41] and using a dedicated negation detection pipeline [42].

Finally, it should be noted that even though SBERT is specifically trained to track semantic similarity, this does not resolve more fundamental theoretical questions concerning semantic representation in LLMs – related to issues such as environmental grounding, reference, or representational format, e.g. [23], [43], [44], [45], [46]. Such matters remain in need of further scrutiny and are beyond the scope of this paper.

On a more philosophical note, it is worth mentioning that “semantic similarity” itself is a problematic concept if interpreted as invoking a distinction between relations that are purely semantically grounded and those that involve further aspects of “world knowledge”. This is close to the highly contentious analytic-synthetic distinction in the philosophy of language. For example, if the embeddings for *dog* and *cat* are close in cosine similarity, does this mean that they share many defining semantic features, or merely that our general world-knowledge assigns many of the same properties to both dogs and cats? The former interpretation requires a commitment to some kinds of semantic definitions underlying word meanings, which has commonly been denied in philosophical, linguistic, and psychological literature [47], [48]. The latter interpretation shuns a sharp distinction between semantic and non-semantic knowledge, which in turn threatens to

TABLE 10. Models tested.

Model	Full name	Base model	Embedding size	Layers	Fine-tuning data	Language
M1	paraphrase-MiniLM-L6-v2	BERT-small	384	6	paraphrase	English
M2	all-mpnet-base-v2	MPNet	768	12	several	English
M3	paraphrase-multilingual-mpnet-base-v2	MPNet	768	12	paraphrase	several (50+)
M4	multi-qa-distilbert-cos-v1	DistilBERT	768	6	question-answer	English
M5	paraphrase-albert-small-v2	ALBERT-small	768	6	paraphrase	English

TABLE 11. Meta-correlations between the observed correlations in human data and correlations in synthetic datasets generated with different models and varying magnitude of noise (σ).

Model	Meta-correlation				
	$\sigma = 0$	$\sigma = 0.5$	$\sigma = 1.0$	$\sigma = 2.0$	$\sigma = 3.0$
M1	0.17	0.33	0.44	0.42	0.39
M2	0.43	0.55	0.63	0.60	0.52
M3	0.58	0.60	0.56	0.46	0.39
M4	0.37	0.36	0.33	0.29	0.23
M5	0.53	0.57	0.49	0.39	0.35

dilute the notion of “semantic similarity” so much that it becomes uninformative.

However, since sentence comparison models are fine-tuned on human-labelled sentence pairs, for our purposes it is sufficient to operationalize “semantic similarity” as simply whatever property those labels capture. This allows using SBERT for obtaining semantic similarity measures in practical applications, without relying on any particular answer to the more philosophical dilemmas.

One concrete way to alleviate the problem would be to reanalyse semantic similarity via a better-understood notion: entailment. Sentence A entails sentence B if A being true requires B being true. The questionnaire analysed provides multiple examples where the entailment can be present. For example, q14r.13 (It would make me proud to own a conditionally automated car.), can be thought to entail the willingness to use such a car. In effect, this would allocate sentence similarity detection in the broader domain of natural language inference.

VI. CONCLUSION

The study presented a novel method to analyse the influence of semantic similarity on survey responses for Likert-scale questionnaires. The method used an LLM-based SBERT algorithm to estimate the semantic similarity matrix of the

response sentences and generated synthetic data using it. Generated synthetic data showed significant similarities to the observed data, collected with a survey from human participants.

Semantically generated synthetic data could be used for piloting questionnaires before data collection. Researchers could test if they can identify the measurement model with their set of questions, and if the semantic similarities might pose a problem for discriminating between factors and interpreting their relationships. This could help in designing better questionnaires and more accurately interpreting the collected data.

Future research should explore the use of the presented method in designing novel questionnaires and in the analysis of a wider range of questionnaires. Future research should try to establish a recommended level of semantic similarity between questionnaire responses according to the measurement model used.

A larger number of LLMs could be tested to optimize the performance of the method. The method could be compared to generating synthetic data via prompting generative LLMs. Research is also needed to further understand why semantic similarity influences survey responses. This could involve empirical experiments with human participants to establish the link between semantic similarity and observed correlations. The concept of semantic similarity itself is also complicated. We have suggested that in the current context, the effect of semantic similarity might be rather interpreted as entailment.

APPENDIX A MODEL SELECTION

Five different models were tested. Three of the models are fine-tuned on paraphrases (M1, M3, M5), one on question-answer pairs (M4), and one on several datasets including title-body, question-answer, and duplicate question pairs (M2). One model (M3) is trained on over 50 languages, while the rest are English-centric. Sentence embeddings are 384-dimensional in M1 and 768-dimensional in the rest. Three models (M1, M4, M5) use different 6-layer derivatives of the BERT architecture as the base model: BERT-small in M1, DistilBERT [49] in M4, and ALBERT [50] in M5. The remaining two models (M2, M3) are built on the 12-layer

TABLE 12. Summary of the first answer.

Parameter	Value
ID of the question to be answered	q12.1
Sentence to be completed	I <option> that I would be suspicious of conditionally automated cars.
Memory contents	None
Response	I strongly agree that I would be suspicious of conditionally automated cars. (5)
Discretized noise	0
Final response	I strongly agree that I would be suspicious of conditionally automated cars. (5)

TABLE 13. Comparison of the response options to the memorized sentence for the second answer.

Sentence in the memory	Cosine similarity	Potential response sentences (numeric coding in parenthesis)
I strongly agree that I would be suspicious of conditionally automated cars. (q12.1)	0.890	I agree that I would trust conditionally automated cars. (4)
	0.888	I strongly agree that I would trust conditionally automated cars. (5)
	0.836	I strongly disagree that I would trust conditionally automated cars. (1)
	0.835	I disagree that I would trust conditionally automated cars. (2)
	0.779	I neither agree nor disagree that I would trust conditionally automated cars. (3)

TABLE 14. Summary of the second answer.

Parameter	Value
ID of the question to be answered	q12.2
Sentence to be completed	I <option> that I would trust conditionally automated cars.
Memory contents	See Appendix Table 4.
Response	I agree that I would trust conditionally automated cars. (4)
Discretized noise	1
Final response	I strongly agree that I would trust conditionally automated cars. (5)

MPNet base model, which combines masked token prediction and perturbed token prediction in pre-training [51]. Information about the five models is summarized in Appendix Table 10. All models were of the main branch available at huggingface.com on 3rd September 2024.

1,000 samples of synthetic data were generated by all the models varying the magnitude of the noise ($\sigma = 0, 0.5, 1.0, 2.0, 3.0$). Between item correlations in the synthetic data were calculated and compared to between item correlations in human data (Appendix Table 11). The strongest of such “meta-correlations” was achieved with M2 using the noise value of 1.0. M3 was the second-best model. M4 was the worst

Four observations can be made from these results. First, larger 12-layer models (M2, M3) fared better than smaller 6-layer ones. This is unsurprising considering the general improvement of model performance when the number of parameters is increased.

Second, fine-tuning with several datasets (M2) was most powerful for capturing the semantic similarities between the full response sentences, but paraphrase-based fine-tuning worked better than M4 fine-tuned with question-answer pairs. This makes sense given that answers contain *new* information not present in the question. Fourth, some noise used in the generation of the synthetic data improved the performance in general, except for M4.

TABLE 15. Comparison of the potential responses to two sentences in the memory for the third answer.

Memorized sentence (ID given in parentheses)	Cosine similarity	Potential response to q15r.2 (numeric coding given in parentheses)
I strongly agree that I would be suspicious of conditionally automated cars. (q12.1)	0.787	I strongly agree that the next car I will buy [...] (5)
	0.778	I agree that that the next car I will buy [...] (4)
	0.755	I strongly disagree that the next car I will buy [...] (1)
	0.744	I neither agree nor disagree that the next car I will buy [...] (3)
	0.722	I disagree that the next car I will buy [...] (2)
I strongly agree that I would trust conditionally automated cars. (q12.2)	0.796	I strongly agree that the next car I will buy [...] (5)
	0.785	I agree that the next car I will buy [...] (4)
	0.752	I neither agree nor disagree the next car I will buy [...] (3)
	0.734	I strongly disagree that the next car I will buy [...] (1)
	0.709	I disagree that the next car I will buy [...] (2)

TABLE 16. Summary of the third answer.

Parameter	Value
ID of the question to be answered	q15.2
Sentence to be completed	I <option> that the next car I buy will be a conditionally automated car, if it is available.
Memory contents	See Appendix Table 6.
Response	I strongly agree that the next car I buy will be a conditionally automated car, if it is available. (5)
Discretized noise	-1
Final response	I agree that the next car I buy will be a conditionally automated car, if it is available. (5)

APPENDIX B
EXAMPLE OF THE RESPONSE PROCESS

The following demonstrates how the response agent answers into the questionnaire. The answering process is the 43rd iteration of the synthetic data used in the analysis.

The first question (q12.1) was answered at random. In this case, the discretized noise was zero and it did not change the response either (Appendix Table 12).

For the second answer, the memory contained one response sentence, which was compared to five potential response sentences to the second question (q12.2) (Appendix Table 13).

The response option “Agree” had the highest cosine similarity to the memorized sentence, but because the discretized

noise value was 1, the response was changed to “Strongly agree” (4 + 1 = 5) (Appendix Table 14).

The sentences q12.1 and q12.2 were rated semantically similar, because both are about trust on automated cars. q12.2 was a negated statement, intended to be reverse coded. It would be more logical that a human respondent who would agree to q12.2 (I would trust...) would disagree to q12.2 (I would be suspicious...). The numerically coded response value would be reverse coded for the analysis, so that the responses would be in line.

Semantic similarity estimation did not recognize negation, and therefore the response agent could agree to both questions. Consequently, the numerically coded responses were

already in line, and we did not apply reverse coding before running CFA or SEM with the synthetic data.

For the third answer, memory contained two response sentences. This means five comparisons in total (Appendix Table 15). Strongly agree to q12.2 has the highest cosine similarity and therefore it is selected as the response. Because the discretized noise value is -1 , the final response option becomes Agree ($5 - 1 = 4$).

The consecutive answers are given in a similar way. The memory always grows by the given answers, and therefore the number of comparisons also grows. The answering stops once all the questions has been answered.

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